

Worked Solutions

Thursday, May 8, 2025 11:24 AM

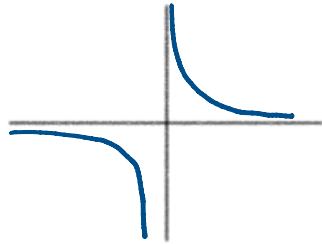
Problem: Sketch isotherms

(a) $T = xy$

Soln Consider $xy = k$ where k is a constant

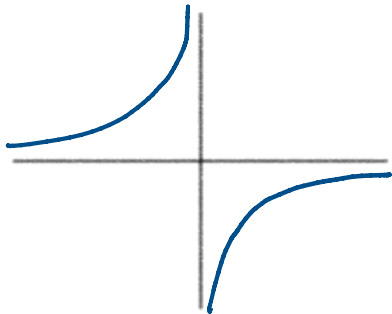
(i) If $k=0$: $xy=0 \Rightarrow x=0$ or $y=0 \rightarrow$ the axes!

(ii) If $k=1$: $xy=1$ is an example of a rectangular hyperbola



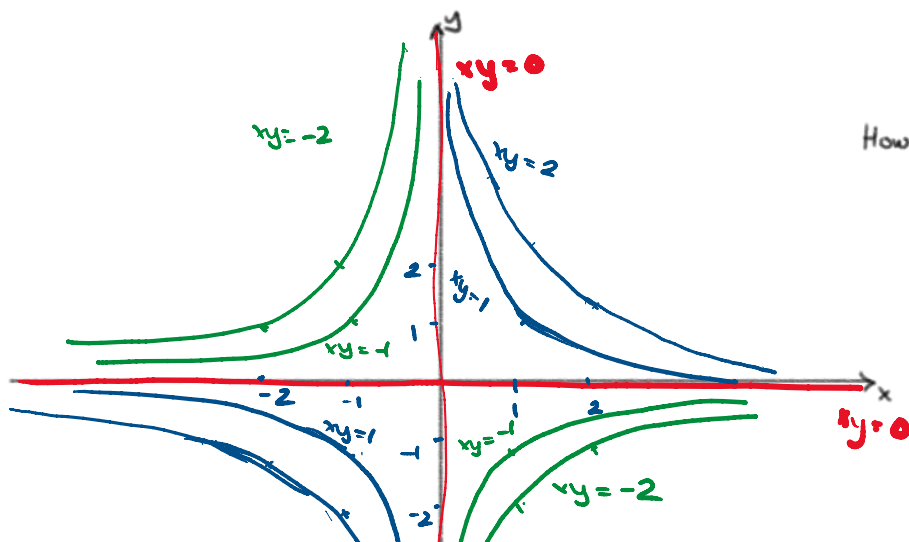
positive values of k look similar

(iii) If $k=-1$, $xy=-1$ is rectangular hyperbola as well, but 'flipped'

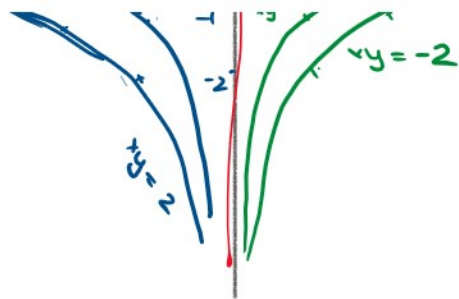


Other negative values of k look similar

Isotherms:



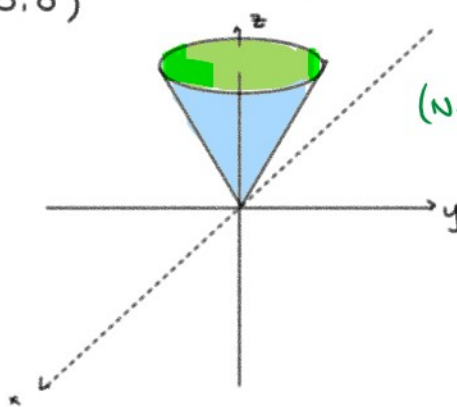
How many do I need to show?



(b) $T = z - \sqrt{x^2 + y^2}$

Soln Consider $z - \sqrt{x^2 + y^2} = k$

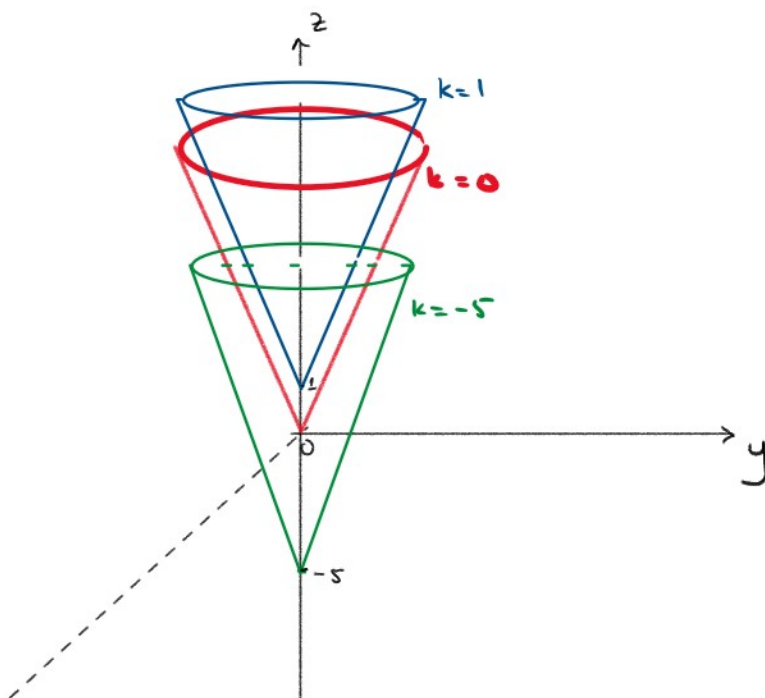
(i) When $k=0$: $z = \sqrt{x^2 + y^2}$ is the upper half of a right circular cone
(How do I know this?)
with vertex at $(0, 0, 0)$



(Note, this is a 2D surface.
inside of the cone is not
part of the isotherm)

(II) For any k : $z - k = \sqrt{x^2 + y^2}$ means we are just "shifting the cone up by k " units

Isotherms:





Problem: Find the first partial derivatives of

$$\vec{v} = [e^x \cos y, e^x \sin y]$$

Soln: Need $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$

$$\frac{\partial}{\partial x} \vec{v} = \begin{bmatrix} \frac{\partial}{\partial x} e^x \cos y \\ \frac{\partial}{\partial x} e^x \sin y \end{bmatrix} = \begin{bmatrix} e^x \cos y \\ e^x \sin y \end{bmatrix}$$

$$\frac{\partial}{\partial y} \vec{v} = \begin{bmatrix} \frac{\partial}{\partial y} e^x \cos y \\ \frac{\partial}{\partial y} e^x \sin y \end{bmatrix} = \begin{bmatrix} e^x (-\sin y) \\ e^x (\cos y) \end{bmatrix} = \begin{bmatrix} -e^x \sin y \\ e^x \cos y \end{bmatrix}$$

Problem: Given $C: \vec{r}(t) = [t, \frac{1}{2}t^2, 1]$

find $\vec{r}'(t)$, a unit tangent vector $\vec{u}$
and a tangent line for C at $P(2, 2, 1)$

Sketch the curve and the tangent

Soln:

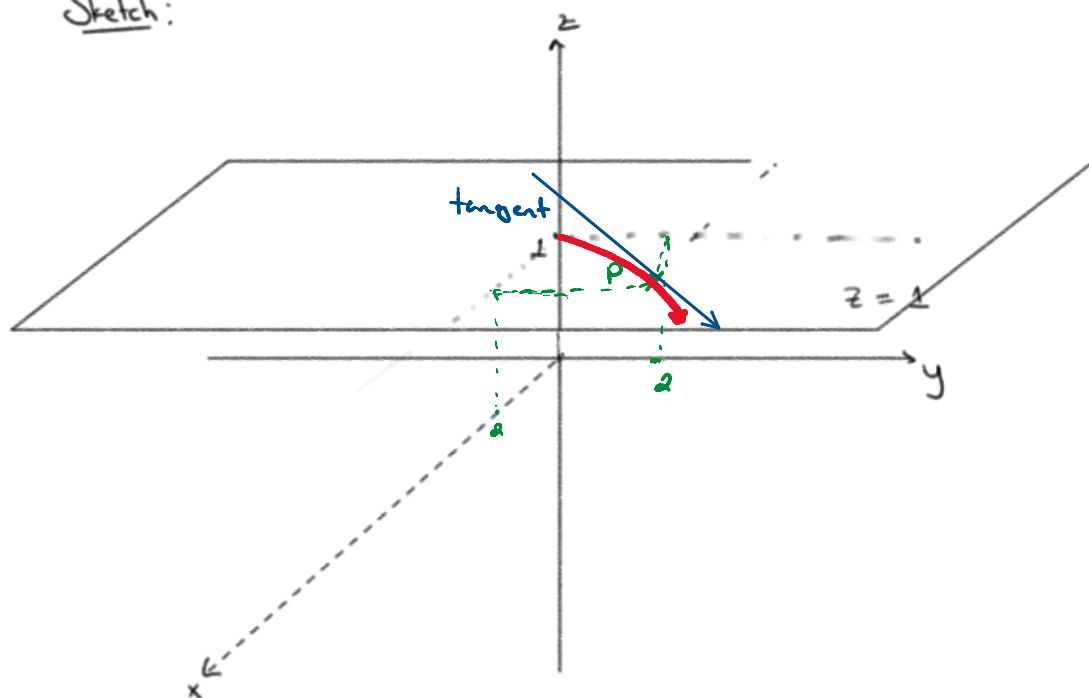
$$\vec{r}'(t) = \begin{bmatrix} 1 \\ t \\ 0 \end{bmatrix}. \quad \text{At } P(2, 2, 1) \text{ we set } \begin{bmatrix} t \\ \frac{1}{2}t^2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \text{ and get } t=2$$

$$\text{So } \vec{r}'(2) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \text{ and } |\vec{r}'(2)| = \sqrt{1^2 + 2^2 + 0^2} = \sqrt{5}$$

$$\text{Thus: } \vec{u} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

tangent-line:
 $\vec{q}(w) = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + w \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad w \in \mathbb{R}$

Sketch:



Problem: For given curve $C: \vec{r}(t) = [t, t^2, 0]$, find $\vec{v}$ and $\vec{a}$, and then decompose $\vec{a}$ into tangential and normal accelerations.

Soln:

$$\vec{v} = \vec{r}'(t) = \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix}$$

$$\vec{a} = \vec{r}''(t) = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$\vec{a}_{\text{tan}} = \left(\frac{\vec{a} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \right) \vec{v} = \frac{\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix}}{\begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix}} \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix} = \frac{4t}{1+4t^2} \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix}$$

$$\begin{aligned} \vec{a}_{\text{norm}} &= \vec{a} - \vec{a}_{\text{tan}} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \frac{4t}{1+4t^2} \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix} \\ &= \frac{1}{1+4t^2} \begin{bmatrix} 0 \\ 2+8t^2 \\ 0 \end{bmatrix} - \frac{4t}{1+4t^2} \begin{bmatrix} 1 \\ 2t \\ 0 \end{bmatrix} \end{aligned}$$

$$= \frac{1}{1+4t^2} \begin{bmatrix} 2+8t^2 \\ 0 \end{bmatrix} - \frac{1}{1+4t^2} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{1+4t^2} \begin{bmatrix} -4t \\ (2+8t^2) - 8t^2 \\ 0 \end{bmatrix} = \frac{1}{1+4t^2} \begin{bmatrix} -4t \\ 2 \\ 0 \end{bmatrix}$$

$$= \frac{2}{1+4t^2} \begin{bmatrix} -2t \\ 1 \\ 0 \end{bmatrix}$$